

Standalone numerical study for the positive-rate gDMR parameter set

Mara Kalicanin Dimitrov¹ and Ying Ni¹

¹Department of Business and Mathematics, Mälardalen University,
721 23, Västerås, Sweden.

1 Numerical studies

This standalone document repeats the manuscript numerical-study design for the positive-rate gDMR parameter set reported in Table 1. The option is a Bermudan put with twelve exercise dates, maturity $T = 1$, and strikes $K = 70, 80, 90, 100, 110$. The direct plain LSMC estimator is compared with the Hybrid LSMC–PDE estimator. All prices, standard errors, confidence intervals, and relative errors are computed from the direct estimators, and benchmark-relative errors use the plain LSMC references in Table 2.

The benchmark references are obtained with 1200 Euler time steps and 1, 200, 000 direct paths. Across the five strikes, the benchmark direct prices range from 0.157901 to 12.571373. These values define the reference scale used in the step-sweep and path-sweep experiments below.

1.1 Fixed path budget, varying number of Euler time steps

The first experiment keeps the direct path budget fixed and varies the number of Euler time steps. The two path budgets are 20, 000 and 60, 000, and both methods are evaluated at 24, 48, 72, and 96 steps. This isolates how the time discretization affects the direct Bermudan estimator away from the benchmark regime.

For 20, 000 paths, the LSMC relative-error range over the full step-sweep grid is 0.069%–6.358%, while the Hybrid LSMC–PDE range is 0.432%–10.872%. Figure 1 shows the strike-wise relative-error profiles, and Table 3 reports the representative 72-step values.

For 60, 000 paths, the same design gives an LSMC relative-error range of 0.177%–2.138% and a Hybrid LSMC–PDE range of 0.491%–11.075%. Figure 2 and Table 4 give the corresponding graphical and tabular summaries.

Table 1 Model and numerical parameters for the standalone positive-rate study.

Quantity	Value
Model	gDMR
Option	Bermudan put
S_0	100.0
K	70, 80, 90, 100, 110
T	1.0
r	0.03
v_0	0.04
v'_0	0.04
κ_1	2.0
κ_2	1.0
θ	0.04
ξ_1	0.35
ξ_2	0.20
ρ_{12}	0.20
ρ_{13}	0.10
ρ_{23}	0.10
δ_1	0.5
δ_2	0.5
Exercise dates	12
Benchmark	LSMC, 1200 steps, 1,200,000 paths
Hybrid asset grid	301

Table 2 Benchmark direct LSMC reference prices for the standalone study.

Strike	Price	SE	CI lower	CI upper
70	0.157901	0.001152	0.155643	0.160159
80	0.741358	0.002648	0.736168	0.746548
90	2.503886	0.004943	2.494198	2.513574
100	6.343052	0.007586	6.328183	6.357921
110	12.571373	0.009593	12.552571	12.590175

1.2 Fixed number of Euler time steps, varying path counts

The second experiment fixes the number of Euler time steps and varies the direct path budget. The reported path grid is 250, 1000, 5000, 10000, 20000, 40000, and 60000. The two discretization levels are 48 and 60 steps.

At 48 steps for the at-the-money strike, the LSMC relative-error range over all path counts is 0.431%–13.884%, while the Hybrid LSMC–PDE range is 0.431%–0.832%. Figure 3 reports the full strike-wise path sweep, and Table 5 reports the 20,000-path representative values.

At 60 steps for the at-the-money strike, the corresponding LSMC range is 0.614%–15.143%, and the Hybrid LSMC–PDE range is 0.312%–1.057%. Figure 4 and Table 6 provide the companion summaries.

The two experiment families provide a direct check of the pricing behavior under the positive-rate parameter set. The tables should be read together with the confidence intervals, since both methods retain finite Monte Carlo variability at the moderate

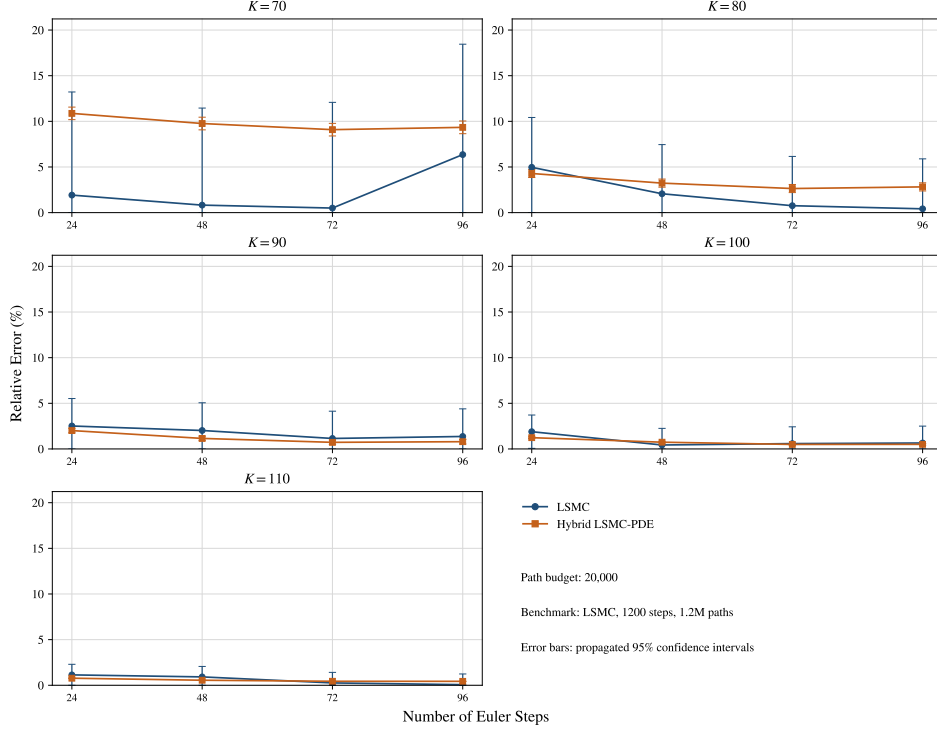


Fig. 1 Relative errors for the matched 20,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 3 Representative direct metrics at 20,000 paths and 72 Euler steps.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	0.158691	0.500%	0.172252	9.089%
80	0.746989	0.760%	0.760926	2.640%
90	2.532665	1.149%	2.522034	0.725%
100	6.380289	0.587%	6.374480	0.495%
110	12.603871	0.259%	12.626499	0.439%

Table 4 Representative direct metrics at 60,000 paths and 72 Euler steps.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	0.159447	0.979%	0.174082	10.248%
80	0.735789	0.751%	0.766830	3.436%
90	2.533891	1.198%	2.533139	1.168%
100	6.401666	0.924%	6.387661	0.703%
110	12.608541	0.296%	12.636272	0.516%

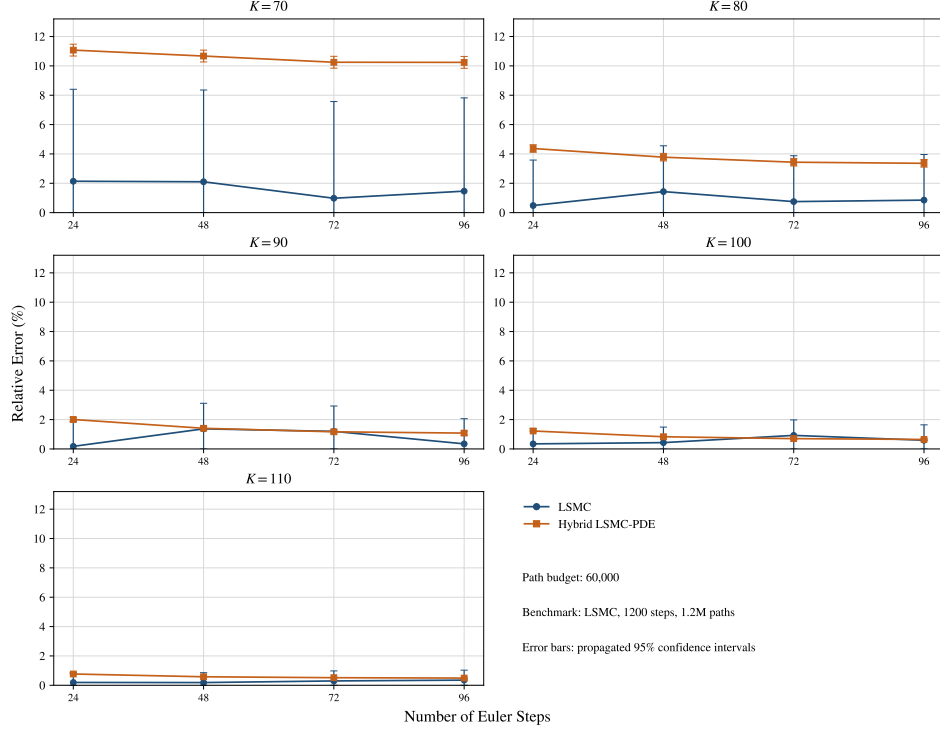


Fig. 2 Relative errors for the matched 60,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 5 Representative direct metrics at 20,000 paths under the fixed 48-step path sweep.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	0.156602	0.823%	0.173313	9.761%
80	0.756684	2.067%	0.765319	3.232%
90	2.554533	2.023%	2.532786	1.154%
100	6.370571	0.434%	6.390036	0.741%
110	12.686787	0.918%	12.640804	0.552%

Table 6 Representative direct metrics at 20,000 paths under the fixed 60-step path sweep.

Strike	LSMC price	LSMC rel. err.	Hybrid price	Hybrid rel. err.
70	0.159514	1.022%	0.172613	9.317%
80	0.756639	2.061%	0.762768	2.888%
90	2.559929	2.238%	2.526756	0.913%
100	6.389520	0.733%	6.381559	0.607%
110	12.633992	0.498%	12.633465	0.494%

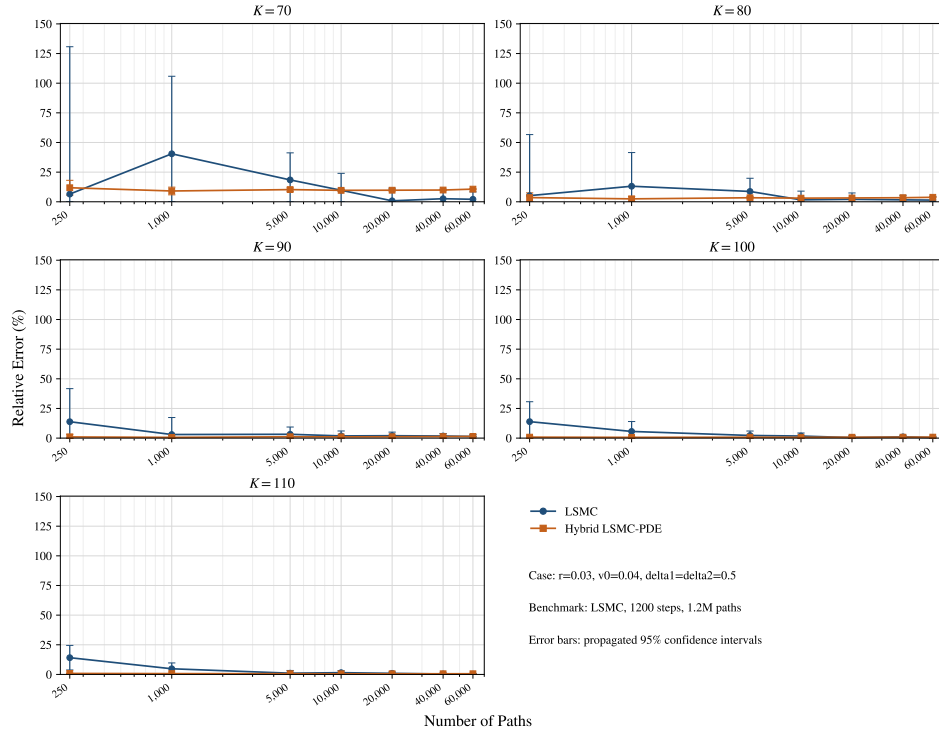


Fig. 3 Relative errors for the fixed 48-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

path budgets used in the sweeps. The appendix-style tables below list the underlying prices used to form the figures and representative summaries.

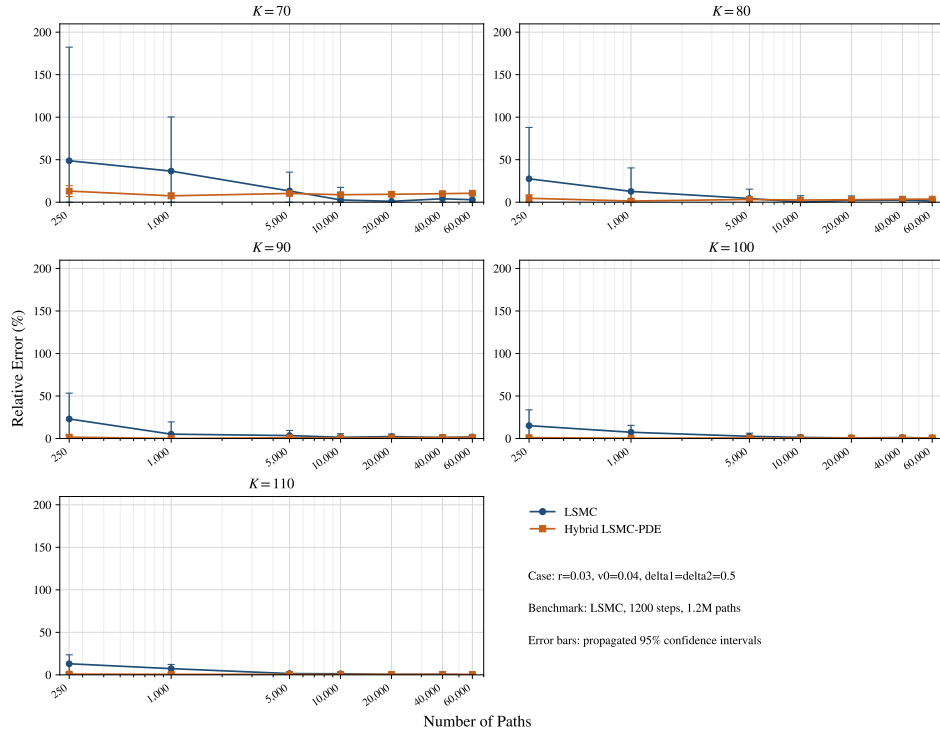


Fig. 4 Relative errors for the fixed 60-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Positive-rate gDMR price tables

Table 7 Benchmark prices. Prices are rounded to three decimals.

Strike	70	80	90	100	110
LSMC benchmark	0.158	0.741	2.504	6.343	12.571

Table 8 Fixed-path step-sweep prices with 20,000 paths. Prices are rounded to three decimals.

Method/step	24	48	72	96
LSMC K=70	0.161	0.157	0.159	0.168
Hybrid LSMC-PDE K=70	0.175	0.173	0.172	0.173
LSMC K=80	0.778	0.757	0.747	0.744
Hybrid LSMC-PDE K=80	0.773	0.765	0.761	0.762
LSMC K=90	2.567	2.555	2.533	2.538
Hybrid LSMC-PDE K=90	2.554	2.533	2.522	2.524
LSMC K=100	6.463	6.371	6.380	6.384
Hybrid LSMC-PDE K=100	6.421	6.390	6.374	6.375
LSMC K=110	12.715	12.687	12.604	12.580
Hybrid LSMC-PDE K=110	12.669	12.641	12.626	12.626

Table 9 Fixed-path step-sweep prices with 60,000 paths. Prices are rounded to three decimals.

Method/step	24	48	72	96
LSMC K=70	0.155	0.161	0.159	0.156
Hybrid LSMC-PDE K=70	0.175	0.175	0.174	0.174
LSMC K=80	0.745	0.752	0.736	0.735
Hybrid LSMC-PDE K=80	0.774	0.769	0.767	0.766
LSMC K=90	2.508	2.538	2.534	2.495
Hybrid LSMC-PDE K=90	2.554	2.539	2.533	2.531
LSMC K=100	6.365	6.370	6.402	6.381
Hybrid LSMC-PDE K=100	6.421	6.396	6.388	6.384
LSMC K=110	12.595	12.548	12.609	12.616
Hybrid LSMC-PDE K=110	12.668	12.644	12.636	12.633

Table 10 Fixed 48-step path-sweep prices. Prices are rounded to three decimals.

Method/strike	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC K=70	0.168	0.222	0.187	0.143	0.157	0.162	0.161
Hybrid LSMC-PDE K=70	0.177	0.172	0.174	0.173	0.173	0.174	0.175
LSMC K=80	0.703	0.839	0.806	0.729	0.757	0.753	0.752
Hybrid LSMC-PDE K=80	0.768	0.760	0.767	0.764	0.765	0.767	0.769
LSMC K=90	2.850	2.580	2.585	2.550	2.555	2.548	2.538
Hybrid LSMC-PDE K=90	2.529	2.519	2.532	2.530	2.533	2.536	2.539
LSMC K=100	7.224	6.700	6.485	6.454	6.371	6.411	6.370
Hybrid LSMC-PDE K=100	6.394	6.383	6.390	6.386	6.390	6.393	6.396
LSMC K=110	14.347	13.171	12.702	12.758	12.687	12.618	12.548
Hybrid LSMC-PDE K=110	12.677	12.659	12.648	12.638	12.641	12.643	12.644

Table 11 Fixed 60-step path-sweep prices. Prices are rounded to three decimals.

Method/strike	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC K=70	0.235	0.216	0.179	0.154	0.160	0.164	0.162
Hybrid LSMC-PDE K=70	0.179	0.170	0.174	0.172	0.173	0.174	0.174
LSMC K=80	0.945	0.836	0.774	0.741	0.757	0.760	0.751
Hybrid LSMC-PDE K=80	0.776	0.752	0.765	0.760	0.763	0.767	0.768
LSMC K=90	3.080	2.633	2.591	2.540	2.560	2.533	2.551
Hybrid LSMC-PDE K=90	2.545	2.504	2.527	2.521	2.527	2.535	2.536
LSMC K=100	7.304	6.810	6.507	6.428	6.390	6.415	6.382
Hybrid LSMC-PDE K=100	6.410	6.363	6.381	6.375	6.382	6.391	6.392
LSMC K=110	14.211	13.488	12.760	12.727	12.634	12.650	12.571
Hybrid LSMC-PDE K=110	12.682	12.643	12.639	12.628	12.633	12.639	12.640